

HARSHA VARDHAN RAPETI

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TECHNICAL SKILLS

Languages C++, Python, Rust, Bash, TypeScript, MySQL, Ruby, Faust
Frameworks FastAPI, Ruby on Rails, JUCE, LangChain, React, TensorFlow
Developer Tools Git, CMake, Make, Visual Studio, GDB, Docker, Valgrind, Perf, Postman

PROFESSIONAL EXPERIENCE

Junior C++ Developer

Aug 2024 – May 2025

Pitch Innovations

- Developed **VST and standalone audio plugins** for Windows and macOS using **C++ and JUCE framework**, deployed to **1,000+ active users**.
- Architected **cross-platform build systems** using CMake, reducing **compilation time by 40%** and enabling concurrent Windows/Mac/Linux builds.
- Integrated **Faust DSP language** into build pipeline, accelerating audio effect **development cycle by 60%** through automated code generation.
- Implemented **8+ professional-grade audio effects** including delays, compressors, distortions, Schroeder reverb, granular synthesis, and convolution reverb with **optimized real-time performance**.

PROJECTS

Warmer - Node-Based Audio Synthesizer [repo](#)

Open-source audio synthesis platform with 25+ unique nodes and real-time graph processing

- Engineered **lock-free multi-threaded audio processing engine** using topological sorting algorithms, supporting **graphs with 100+ nodes** at 48kHz sample rate
- Implemented **15+ DSP algorithms** including FFT-based spectral processing, IIR/FIR digital filters, and non-linear waveshapers.
- Designed efficient **XML-based serialization system** for graph state persistence.
- Technologies: C++17, CMake, Make, Mingw-w64, JUCE, GDB, Valgrind, Git

Analytiks - Audio Visualization Software [repo](#)

May 2025 – Jun 2025

Professional audio analysis plugin with OpenGL-accelerated visualizers

- Developed **4 real-time audio visualizers** (Correlation Meter, Spectrogram, Oscilloscope, Spectrum Analyzer) using OpenGL shaders, achieving **60+ FPS at 1080p resolution**
- Optimized inter-thread communication using **non-blocking lock-free queues**.
- Automated build and distribution pipeline using GitHub Actions CI/CD.
- Technologies: C++, CMake, Visual Studio 2022, JUCE, OpenGL, Git, GitHub Actions

SynthBite16 - 16-bit Computer Architecture [repo](#)

Nov 2023 – Jan 2024

Fully functional 16-bit computer design with custom instruction set architecture

- Designed **complete 16-bit computer architecture** with 38 instructions, 2 addressing modes, and hardware register stack in Logisim
- Optimized instruction format using 4-nibble encoding scheme, achieving **95% opcode utilization efficiency**
- Implemented **Harvard architecture** with separate 132KB instruction and data memory spaces
- Created **custom assembly language and assembler in C++**, supporting symbolic addressing
- Technologies: Logisim, C++, Assembly Language Design, Computer Architecture

EDUCATION

BIT Mesra (Birla Institute of Technology)

Oct 2022 – Present

Bachelor of Technology in Artificial Intelligence and Machine Learning
CGPA: 7.58/10.0

PRESENTATIONS AND TALKS

ADC India 2025: Spatial Audio Localization Techniques

Conference Presentation

Presented at Audio Developer Conference (ADCx) India [Watch](#)

- Delivered technical presentation on implementing real-time binaural audio synthesis using Head-Related Transfer Functions (HRTFs)
- Presented to audience of **100+ audio developers, engineers, and industry professionals**